

01 / WHERE THE TOKENS WENT

Supervising the engine cost more than running it.

Between 7 and 16 September the brain campaign's models wrote 9.49M output tokens. The engine's own provider calls account for 2.67M of them. Supervisors working outside the loop — diagnosing, repairing and relaunching the engine — wrote 6.82M.

12%

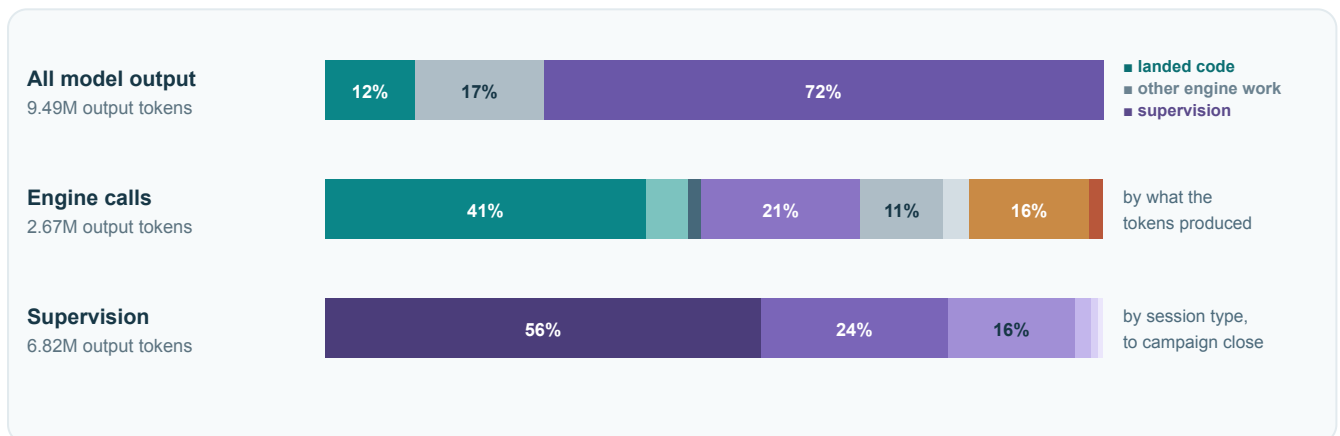
of all output tokens became code that reached the integration branch (1.10M).

2.6×

supervision output relative to engine output; 4.1×

1.6%

of engine output went to in-loop planning, deciding and probes.



Code that landed Audits of landed work Planning · deciding · probes Engine self-repair procedures Code that never landed
Audits of unlanded work Lost: limits · timeouts · kills Lost: invalid verdicts
Codex interactive supervisor Supervisor-launched recovery runs Claude interactive supervisor Supervisor review calls
Grok worker experiments Probe and continuation sessions

WHAT COUNTS AS LANDED

A call's output counts as landed when a commit from its attempt is an ancestor of `codex/brain-integration`, whatever the runner reported. Adopted and successor candidates count. Work that was accepted but never merged does not.

THREE KINDS OF TOKEN, NEVER SUMMED AS WORK

Output is what models wrote. **Fresh input** is new prompt material. **Cached input** is context re-read at a discount — 97% of engine input — and is reported separately, never as work or money.

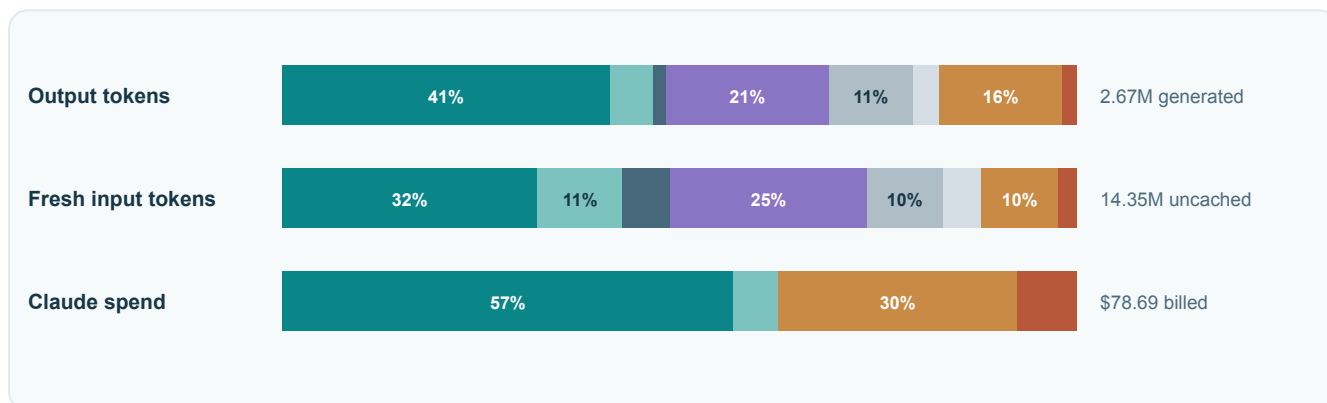
The problem moved. In the 0.20 AXON campaign, planners and critics made half of all engine calls. Here, genuine planning made 8 of 166, and planning, deciding and probes together wrote under 2% of engine output. The overhead now sits outside the loop, in supervisor sessions that no engine metric records.

Engine: 166 calls from `runner.completed` events; 15 calls with no recorded usage were recovered from provider session logs; 2 remain unknown. Supervision: 87 supervisor-launched exec calls, review envelopes and supervisor transcripts, cut at the 0.23.2 release (16 Sep 10:11Z). Claude dollars come from provider envelopes; Codex ran on quota.

02 / INSIDE THE LOOP

The engine's own bureaucracy is small. Its failures are not.

Two-thirds of engine output went to implementation, and deterministic policy made most control decisions without a model. The waste sits in three places: work the engine performed on itself, environment failures, and work that never landed.



Engine calls only. The same outcomes, weighed three ways: what models wrote, what fresh context they read, and what Claude billed.

ENGINE CALL CATEGORY	CALLS	OUTPUT	FRESH INPUT	CACHED	CLAUDE \$
Implementation	56	1.78M 66.9%	7.29M	328M	\$68.43
Engine self-repair procedures	53	548k 20.5%	3.56M	54M	—
Audit	39	295k 11.1%	2.65M	28M	\$10.26
Planning (planner + critic)	8	41k 1.5%	771k	6M	—
Deciding	2	298 0.0%	24k	19k	—
Provider probes	8	63 0.0%	62k	95k	—

CONTROL IS NEARLY FREE

20 of 24 failure decisions went through the deterministic fast path with no model call. 1,774 reconcile cycles ran entirely in code.

SELF-REPAIR IS NOT

On 7–10 September, 53 calls built recovery controllers, proof fixtures and audit bridges for the engine itself: 21% of output. They ran under the planner role, so role metrics count them as planning.

BEFORE AND AFTER 0.21

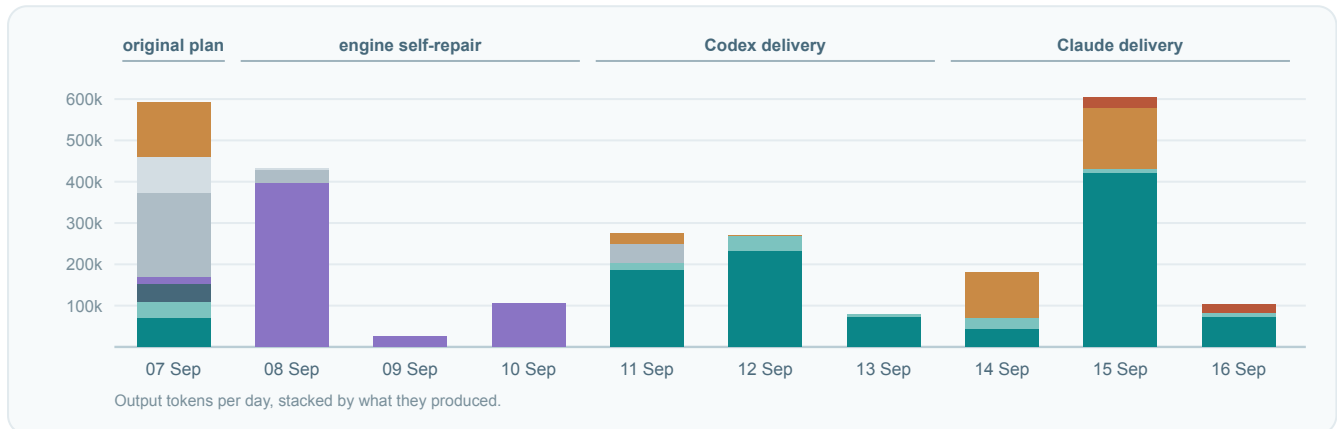
AXON (0.20): planners and critics made 430 of 855 calls. Here those roles made 41 of 166, and 33 were the repair procedures. Planning proper: 8 calls. Different project, so this is not a controlled comparison.

Method. Engine usage comes from `runner.completed` events, never overwritten sidecars. Provider semantics are normalised: Codex input includes cached tokens; Claude cache writes count as fresh input. Landing is tested with `git merge-base --is-ancestor`, including successor and adopted commits named by run ID. The Codex supervisor total sums per-event counter increases, handling resets and forked subagents. Claude sessions are de-duplicated by message ID. Supervision stops at the 0.23.2 release, so this report's own analysis is excluded.

03 / ASSURANCE AND LOSS

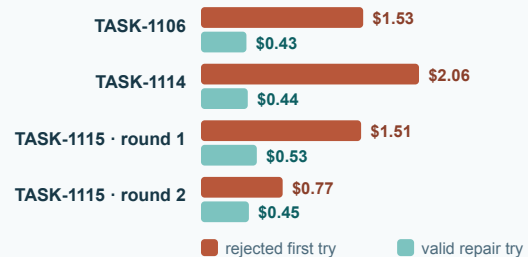
Audits earn their keep. One malformed field wastes most of their dollars.

Auditing took 11% of engine output and returned needs-fix on 15 of 31 valid verdicts. On Claude, a single prompt defect made every audit since 15 September run twice.



Since 15 September, every first audit try fails

All four rejections carry the same host error: `findingsStatus` must be an `object`. The prompt names the field as optional without giving its shape. The model guesses wrong, the host rejects the verdict, and a repair try with the error attached passes. Rejected tries cost \$5.86, which is 57% of Claude audit spend and 3.2× the valid tries.



Where build tokens were lost

Limits, timeouts and kills cost 10 calls, 414k output and \$23.69 of Claude spend with nothing landed. Two one-hour clocks timed out TASK-1114 with no source edits (\$15.15). A session limit killed a TASK-1106 worker (\$8.54). A TASK-1115 worker killed the same way was salvaged, and its code landed. Codex timeouts recorded no usage; their session logs show 158k output tokens that never produced a commit.

WORK STREAM	CALLS	OUTPUT	BUILD OUTPUT THAT LANDED	CLAUDE \$
B1 package + ingest	2	35k	100%	—
B2 context service	5	103k	52%	—
B3 invocation coverage	7	149k	100%	—
B4 reviewed memory	11	164k	100%	\$7.70
B5 evaluation harness	5	242k	57%	\$26.12
Reconciliation discovery (0.23.0)	7	385k	60%	\$22.95
Invocation admission (0.23.1)	9	178k	100%	\$21.92
Brain-package maintenance	18	257k	89%	—
Original plan, 7–8 Sep (abandoned at pivot)	31	567k	16%	—

04 / OUTSIDE THE LOOP

Supervision is the number to drive down.

Supervisors only spent these tokens because the engine could not finish unattended. They produced real fixes — six lifecycle deadlocks diagnosed, runtimes A12–A16 built, releases 0.22.0–0.23.2 shipped — but none of it is the product the campaign set out to build.

SUPERVISION LAYER	FRESH INPUT	CACHED INPUT	OUTPUT
Codex interactive supervisor 7–14 Sep · 3 threads, 23 subagents	28.31M	947M	3.83M 56%
Supervisor-launched recovery runs 12–13 Sep · 87 codex exec calls	10.97M	323M	1.63M 24%
Claude interactive supervisor 13–16 Sep · 2 sessions, to close	6.50M	432M	1.12M 16%
Supervisor review calls claude -p · 6 calls · \$18.14	600k	17M	137k 2%
Grok worker experiments 13 Sep · 2 workers, 1 unrecorded · input not split	13.13M	0	66k 1%
Probe and continuation sessions 5 transcript-only sessions	0	5M	42k 1%
Supervision total	59.51M	1.72B	6.82M
Engine total, for comparison	14.35M	416M	2.67M

What to change next

- 1. Measure supervision, not only the engine.** Report supervisor tokens per integration beside engine tokens per integration. Today no engine metric sees 72% of output.
- 2. Label engine self-repair as its own role.** Its 53 calls ran as “planner”. Also fix `singular metrics`, which sums raw input across providers so Claude calls barely register.
- 3. Stop retrying into provider limits.** Check session headroom before a long implementer call and back off after a 429. The engine retried immediately into the same limit.
- 4. Size worker clocks from the contract** and keep the partial transcript on timeout. Both TASK-1114 timeouts left nothing to resume.
- 5. Fix the auditor’s first try.** Give `findingsStatus` its exact shape in the prompt, or omit it from first-round audits: about \$1.47 and 12k output tokens saved per audited task.

ESTABLISHED

Where the recorded tokens went, which engine output reached the branch, and the ratio of supervision to engine tokens in this campaign.

NOT ESTABLISHED

Whether this much supervision was avoidable at this engine version. The dollar cost of Codex (quota) or of interactive Claude sessions (subscription). Anything beyond this one campaign.